ArgOS

AI-native VC Operating System

Sourcing → Screening → Diligence → Decision

Discover early

Founders surfaced from their public footprint — before they hit startup databases.

Trust, computed

Noisy signals resolved into corroborated claims with auditable, per-claim trust scores.

Three axes, no averaging

Founder · Market · Idea-vs-market scored independently — disagreement is the product.

Stack

Deterministic math in the core, LLMs only at the edges of judgment.

Backend

FastAPI
SQLAlchemy + Alembic
APScheduler — cron jobs
PyMuPDF — deck parsing

Agents

LangGraph + LangChain
Tavily web search
traces in Postgres + vector DB

Data

Postgres + pgvector
kNN claim matching

Frontend

Next.js + TypeScript

Contract

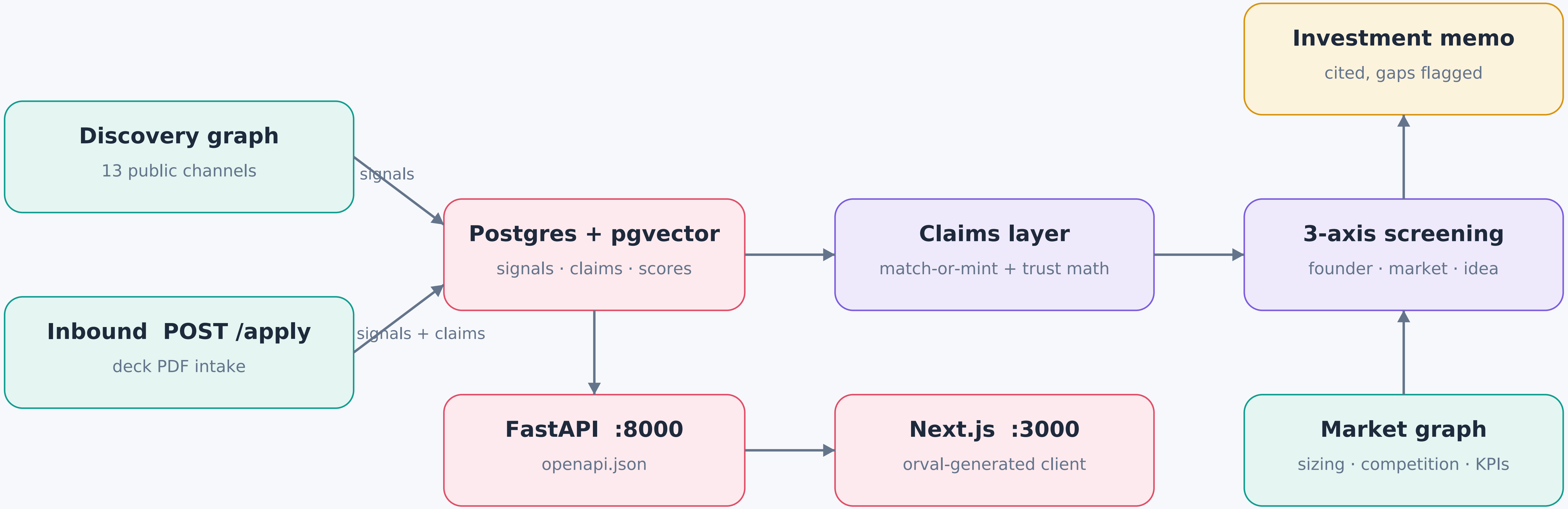
Pydantic → OpenAPI → typed client
one schema, end to end

Quality

pytest on the real DB
ruff + pyright

Architecture

Two LangGraph agents feed one Postgres; claims are minted or matched, screened on three axes, and end in a memo.



Discovery: `plan_searches → run_searches → extract_candidates → research_candidates → finalize (parallel Tavily fan-out)`

Market: `plan_research → run_searches → extract{sizing, competition, comparables, kpi} → synthesize`

Inbound: `deck → per-page signals → one extraction call → deterministic-then-LLM prescreen → screening (~10–20 s)`

Implementation

Every number on screen is auditable — LLMs judge, formulas score.

Deterministic trust math — per claim

Founder Score: saturating exponential, 18-month recency half-life

$$\text{trust} = (1 - \Pi(1 - \text{support})) \cdot \Pi(1 - \text{refute})$$

corroboration saturates toward 1 (noisy-OR) · refutation multiplies toward 0 · weight = reliability × resolution confidence × relevance

LLM only where judgment lives

LLMs: claim adjudication, idea axis, memo prose.
Everything else: deterministic formulas.
Memo citations must resolve to real claims
or they are dropped and flagged as gaps.

Honest market figures

Every figure carries a basis:
reported, bottom-up estimate, or gap.
A flagged gap beats an invented number.

Single-writer persistence

Graph compute decoupled from DB writes:
one JobRun-wrapped writer per run.
Scheduler always on: discovery hourly,
refresh every 6 h.

Anti-hallucination by construction

A claim minted without a supporting
signal citation is rejected.
Frozen eval bed of real pipeline-minted
founders keeps iterations comparable.